Concept Learning

2

Pre

CS 3244
Machine Learning

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Recap and Forecast

Learning is used when

1. A pattern exists
2. We cannot pin it down mathematically
3. We have data on it

Focus on supervised learning:

- Unknown target function

$$y = f(\mathbf{x})$$

- Data set

$$\left(\mathbf{x}^{(1)}, y^{(1)}\right), \ldots, \left(\mathbf{x}^{(m)}, y^{(m)}\right)$$

- Learning algorithm picks $h_\theta \approx f$ from hypothesis set $\mathcal{H}$

Topics for this week:

- Learning from Examples
- General-to-Specific ordering over hypotheses spaces
- Version spaces
- Candidate Elimination Algorithm
- Picking new examples
- The need for inductive bias

Kopi & Teh

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Lim Kopi?

lIm kopi: [leem-koh-pee] (*adjective)* Chill-out, and to relax. Traditionally means to ‘drink coffee’, but has evolved to become synonymous with asking someone out for a drink, or a chat.

Example: “It’s stifling here in the office on this Monday morning, let’s lim kopi at the café and discuss further?” From Hokkien/Malay.

形容词： 放松、悠闲，休息。 原译为”喝咖啡“，但之后演变为约人出去聊天。

口语例句：（源自福建话、马来语）“星期一的办公室实在是令人窒息，不如，我们一块儿去外面lim kopi, 继续讨论我们的议题？”

adjektif. Lepak, berehat. Maksud asalnya ialah ‘minum kopi’, tetapi telah berubah untuk bermaksud mengajak sesorang untuk keluar dan minum bersama-sama.

Contoh bahasa percakapan: “Saya berasa rimas duduk di pejabat pada pagi Isnin, mari lim kopi dan teruskan perbincangan?” Daripada bahasa Hokkien atau Melayu.

லிம் கோப்பி• வினைச்சொல்: சிலுசிலுப்பு, ஓய்வெடுக்க, பொதுவாக கோபி அருந்துதல் என பொருள், ஆனால் நாளடைவில் அது ஒருவரை கோபி அருந்த அழைப்பு விடுவதுபோல் அமைந்து விட்டது.

பேச்சு வழக்கு உதாரணம்: இன்றைய திங்கள் காலை ரொம்ப கடினமாக இருக்கிறது, ஒரு “லிம் கோப்பி” அடித்துகொண்டு உரையாடுவோமா?

ஹோக்கியேன் மொழியிலிருந்து வந்த சொல்.

Credits: Zinkie Aw Singapore Lang Rocks . Logo and text used with explicit permission http://singaporelang.rocks/glossary/lim-kopi

Kopi the local way

Sweet
Local
Coffee

Kopi

Sweet
Local
Tea

Teh

Coffee,
No milk,
No sugar

Kopi O
Kosong

Coffee,
No milk,
Less strong,
No sugar

Kopi O
Po Kosong

Coffee
With milk,
No sugar

Kopi Si
Kosong

Coffee,
Sweeter

Kopi
Ga Dai

Iced
Coffee

Kopi
Peng

Iced Coffee,
To go

Kopi Peng
Tabao

Pictures drawn by Min-Yen Kan. CC BY 4.0

EnjoyDrink Concept

Positive (*aka* +ve, +, 1, 1, True)

Negative (*aka* –ve, –, –1, 0, False)

Concept Learning

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Why Study Concept Learning?

Simple (e.g., assumes error-free, noise-free data) and practically useless!

Easier to fully understand and explain the general challenges, issues, and concepts in ML

White-box model: Prediction is interpretable and explainable

Relates well to your earlier modules: Discrete Math (CS1231), and principles of logic and search in Foundations of AI (CS3243)

Principled and rigorously grounded: Proofs

We'll be following Mitchell's *Machine Learning,* Chapter 2 for this unit

So what is Concept Learning?

Concept ≡ A Boolean-valued function f over a set of instances $\mathbf{X}$ (each $\mathbf{x}$ comprising of discrete input attributes $x_1, x_2, \ldots, x_n$).

Everything is discrete

Relating to Paradigms of Learning: Concept learning is a form of supervised learning!

We infer an unknown Boolean-valued function h (aka $\hat{f}$) from training examples.

S/N	Drink	Milk* **	Sugar* **	Iced	Strength*	For Here	*EnjoyDrink*
1	Kopi (Coffee)	Si (Evap. Milk)	Kosong ("Zero", No)	Peng (Ice)	Gau (Strong)	Lim (Drink)	Yes
2	Kopi	Si	Ga Dai (Extra)	Peng	Gau	Lim	Yes
3	Teh (Tea)	O ("Black")	Ga Dai	Peng	Gau	Tabao (Take Away)	No
4	Kopi	Si	Ga Dai	Peng	Por (Weak)	Tabao	Yes

*: *It's more complicated*

**: *There are interactions. This factorization is not right; just approximate.*

How to Represent a Hypothesis?

Many possible hypothesis representations: Trade-off between expressive power vs. smaller hypothesis space

Consider a simple representation: Hypothesis h is a conjunction of constraints on input attributes

Each constraint can be:

- a specific value (e.g., "*Strength?* = *Gau (Strong)*")
- don't care (e.g., "*Strength?* = *?*")
- no value allowed (e.g., "*Strength?* = $\emptyset$")

Drink	*Milk*	*Sugar*	*Iced*	*Strength*	*For Here*
$\langle ?,$	$Si,$	$?,$	$Peng,$	$?,$	$Lim\rangle$

EnjoyDrink Concept Learning Task

Given:

1. Input instances $\mathbf{X}$: Each instance $\mathbf{x} \in \mathbf{X}$ is represented by the following input attributes describing the drink:
   - *Drink?* (with values *Kopi* and *Teh*)
   - *Milk?* (with possible values *Si* "*Evaporated*", – "*Condensed*", and O "*Black*")
   - *Sugar?* (with values *Kosong* "*No Sugar*" and *Ga Dai* "*Add Sugar*")
   - *Iced?* (with values *Peng* "*Ice*" and –)
   - *Strength?* (with values *Gau* "*Strong*" and *Por* "*Weak*")
   - *For Here?* (with values *Lim* "*Drink*" and *Tabao* "*Take Away*")

EnjoyDrink Concept Learning Task

Given (continued):

2. Hypothesis space $\mathcal{H}$: Each hypothesis $h \in \mathcal{H}$ $(h : \mathcal{X} \rightarrow \{0, 1\})$ is represented by some constraints on input attributes

   (e.g., conjunctive constraints leading to a form like $\langle ?, Si, ?, Peng, ?, Lim \rangle$)

Script notation (e.g., $\mathcal{X}$) ≡ "the space of all possible __"

Definition. An input instance $\mathbf{x} \in \mathcal{X}$ **satisfies** (all of the constraints of) a hypothesis $h \in \mathcal{H}$ iff $h(\mathbf{x}) = 1$.

In other words, h classifies $\mathbf{x}$ as a +ve example.

EnjoyDrink Concept Learning Task

Given (continued):

3. Unknown target concept (function) *EnjoyDrink*: $f: \mathcal{X} \rightarrow \{0, 1\}$

Originally, f denoted as c for "concept"

4. Noise-free training examples $\mathbf{D}$: +ve and −ve examples of the target function (e.g., $\langle \mathbf{x}^{(1)}, f(\mathbf{x}^{(1)}) \rangle, \ldots, \langle \mathbf{x}^{(m)}, f(\mathbf{x}^{(m)}) \rangle$.)

Determine a hypothesis $h \in \mathcal{H}$ that is **consistent** with $\mathbf{D}$.

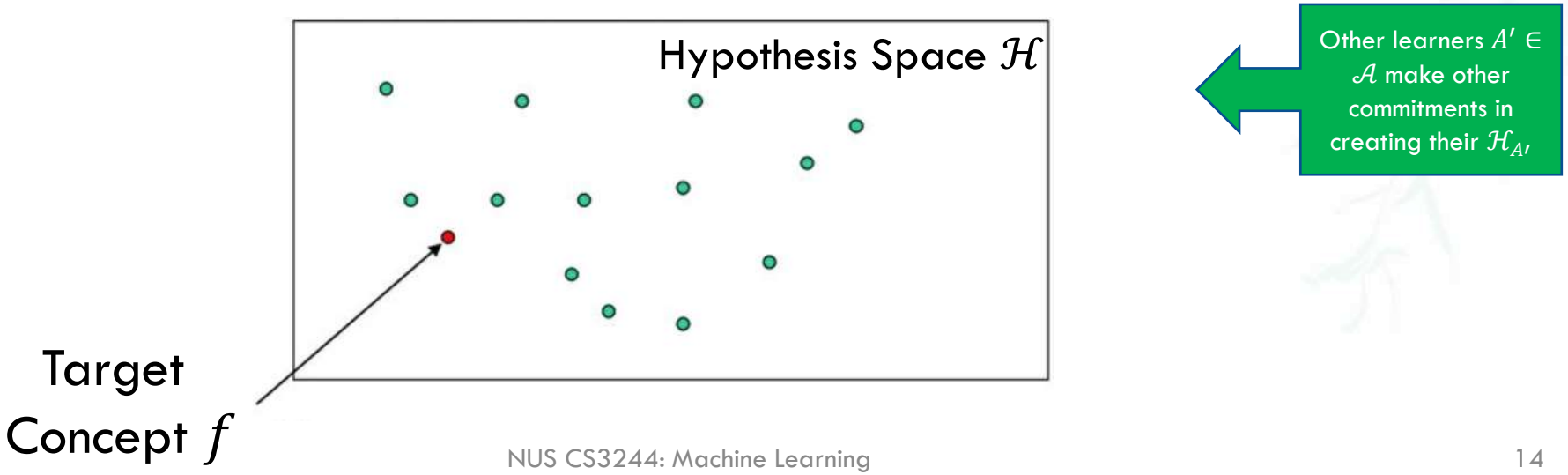
Definition. A hypothesis h is **consistent** with a set of training examples $\mathbf{D}$ iff $h(\mathbf{x}) = f(\mathbf{x})$ for all $\langle \mathbf{x}, f(\mathbf{x}) \rangle \in \mathbf{D}$.

Your Turn: How is saying "h is *consistent* with training example $\langle \mathbf{x}, f(\mathbf{x}) \rangle$" different from saying "instance $\mathbf{x}$ *satisfies* (last slide) h"? What's the implication?

Conjunctive Hypotheses

Let's explore the space of "all conjunctions of constraints" $\leftarrow \mathcal{H}_A$

Our conjunctive learning algorithm A assumes the target concept f falls within this space



Inductive Learning Assumption

A hypothesis close to the target concept f obtained after seeing many training examples will results in high accuracy on a set of unobserved examples.

Training Set $\mathbf{D}$ → Testing Set $\mathbf{D'}$

Hypothesis h approximates f well

Hypothesis h also approximates f well

Inductive Learning Assumption

The learning task is to determine h identical to f
over the entire set of possible instances $\mathcal{X}$.

But the only information about f is its value over $\mathbf{X}$ (training set).

Inductive learning algorithms can at best guarantee that the induced h fits f over $\mathbf{D}$.

Inductive Learning Assumption: Any hypothesis found to approximate the target function well over a sufficiently large set of training examples will also approximate the target function well over other unobserved (test) examples.

Feasibility of Learning

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An Unbiased Learner

Intuition. Choose $\mathcal{H}$ that can express every teachable concept (i.e., $\mathcal{H}$ is the power set of $\mathcal{X}$)

Then we'll need training examples for every possible input instance in $\mathcal{X}$ to converge to the target concept.

Limitation. Cannot classify new unobserved input instances (a.k.a. generalize beyond observed training examples)

Futility of Bias Free Learning

A learner that makes no *a priori* assumptions on f has *no rational basis* for classifying any unseen instances.

Inductive Bias (the prior assumptions of A) is the only way to make learning feasible.

We can measure the amount of bias in a learner, thus ordering them by their generalization power.

Concept Learning as Search

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Concept Learning is Search

Goal. Search for a hypothesis $h \in \mathcal{H}$ that is *consistent* with $\mathbf{D}$.

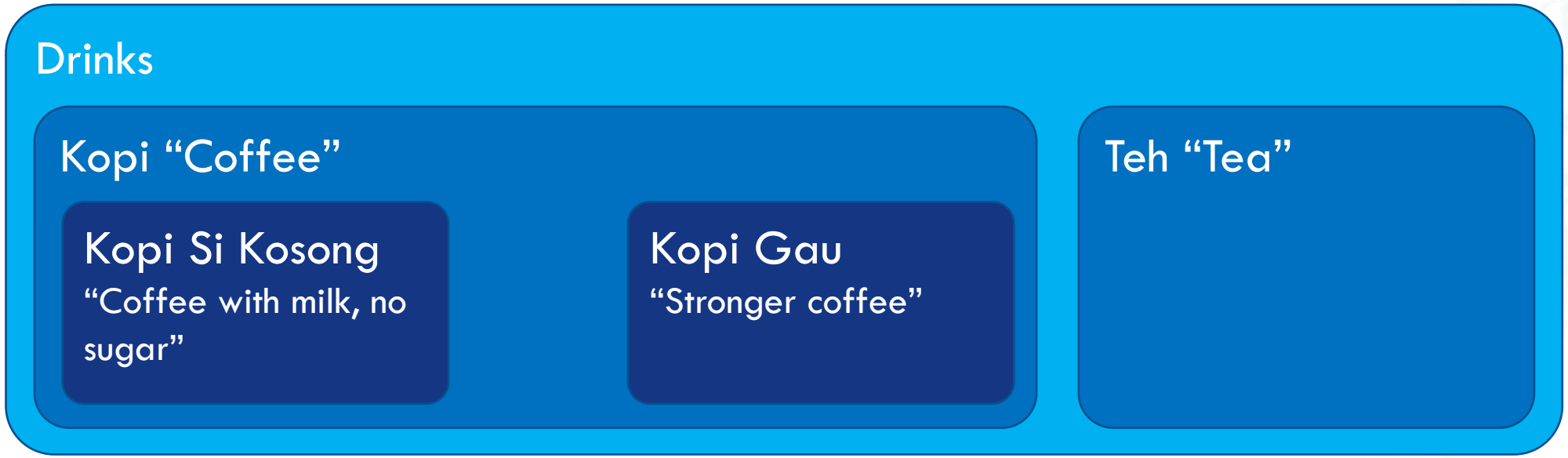
For the *EnjoyDrink* task, $\mathcal{H}$ contains:

- $4 \times 5 \times 4 \times 4 \times 4 \times 4 \; = \; 5120$ synthetically distinct hypotheses
- $1 + 3 \times 4 \times 3 \times 3 \times 3 \times 3 \; = \; 973$ semantically distinct hypotheses

Every hypothesis containing 1 or more ∅ symbols represents an empty set of input instances, hence classifying every instance as a negative example.

In practice, hypothesis space $\mathcal{H}$ is much larger and possibly infinite, hence motivating the need to exploit structure for searching efficiently.

Concept / Hypothesis Generality



A concept p is more general than or equal to another concept q if the set of instances represented by p includes the set of instances represent by q.

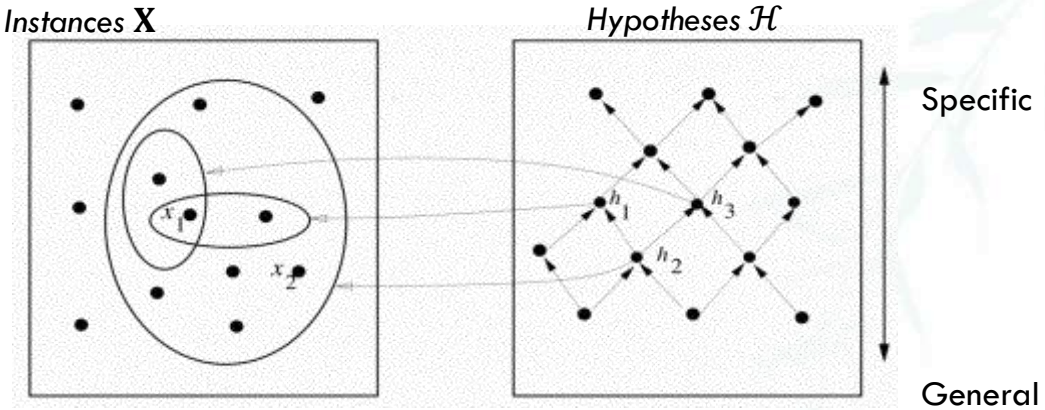
Exploiting Structure in Concept Learning

Definition. h_j is more general than or equal to h_k (denoted by $h_j \geq_g h_k$) iff any input instance $\mathbf{x}$ that satisfies h_k also satisfies h_j:

$$\forall \mathbf{x} \in \mathbf{X} \quad (h_k(\mathbf{x}) = 1) \rightarrow (h_j(\mathbf{x}) = 1).$$

$\geq_g$ relation defines a partial order (reflexive, antisymmetric, transitive) over $\mathcal{H}$ & not a total order.

(e.g., $h_1 \ngeq_g h_3$ and $h_3 \ngeq_g h_1$)



$\mathbf{x}^{(1)}$ = <Kopi, Si, Ga Dai, Peng, Por, Lim>
$\mathbf{x}^{(2)}$ = <Kopi, Si, Ga Dai, –, Gau, Lim>

h_1 = <Kopi, ?, ?, Peng, ?, ?>
h_2 = <Kopi, ?, ?, ?, ?, ?>
h_3 = <Kopi, ?, ?, ?, Por, ?>

Exploiting Structure in Concept Learning

Definition. h_j is more general than or equal to h_k (denoted by $h_j \geq_g h_k$) iff any input instance $\mathbf{x}$ that satisfies h_k also satisfies h_j :

$$h_j \geq_g h_k \equiv \forall \mathbf{x} \in \mathbf{X} \, (h_k(x) = 1) \rightarrow (h_j(\mathbf{x}) = 1) \, .$$

Definition. h_j is more general than h_k (denoted by $h_j >_g h_k$) iff $h_j \geq_g h_k$ and $h_k \ngeq_g h_j$.

Definition. h_j is more specific than h_k iff h_k is more general than h_j.

Find-S

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Find-S Algorithm

Intuition. Start with most specific hypothesis. Whenever it wrongly classifies a +ve training example as −ve, "minimally" generalize it to satisfy its input instance.

1. Initialize h to most specific hypothesis in $\mathcal{H}$
2. For each positive training instance $\mathbf{x}$
   - For each attribute constraint a_i in h

     If $\mathbf{x}$ satisfies constraint a_i in h,

     then do nothing

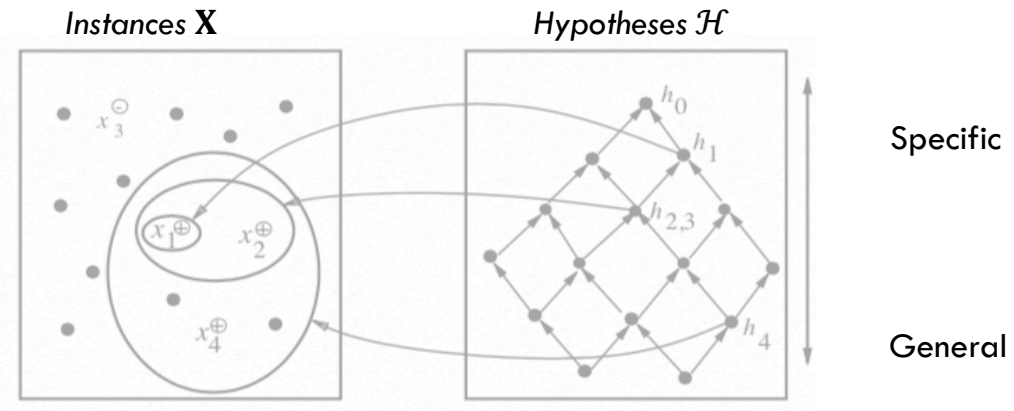
     Else

     replace a_i in h by the next more general constraint that is satisfied by $\mathbf{x}$
3. Output hypothesis h

Finding a maximally specific hypothesis

Hypothesis Space Search by Find-S

Proposition 1. h is consistent with $\mathbf{D}$ iff every +ve training instance satisfies h and every −ve training instance does not satisfy h.

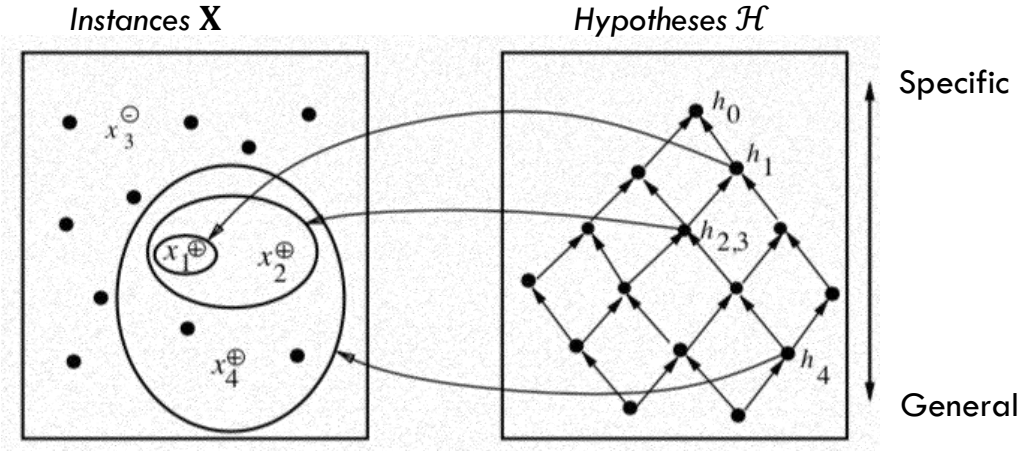


$x^{(1)}$ = <Kopi, Si, Kosong, Peng, Gau, Lim>, +
$x^{(2)}$ = <Kopi, Si, Ga Dai, Peng, Gau, Lim>, +
$x^{(3)}$ = <Teh, O, Gai Dai, Peng, Gau, Tabao>, −
$x^{(4)}$ = <Kopi, Si, Ga Dai, Peng, Por, Tabao>, +

h_0 =<∅,∅,∅, ∅,∅,∅>
h_1 = <Kopi, Si, Kosong, Peng, Gau, Lim>
h_2 = <Kopi, Si, ?, Peng, Gau, Lim>
h_3 = <Kopi, Si, ?, Peng, Gau, Lim>
h_4 = <Kopi, Si, ?, Peng, ? ,? >

Hypothesis Space Search by Find-S

Proposition 2. Suppose that $f \in \mathcal{H}$. Then, h_m is consistent with $\mathbf{D} = \left\{\mathbf{x}^{(j)}, f\left(\mathbf{x}^{(j)}\right)\right\}_{j=1,\ldots,m}$



$x^{(1)}$ = <Kopi, Si, Kosong, Peng, Gau, Lim>, +

$x^{(2)}$ = <Kopi, Si, Ga Dai, Peng, Gau, Lim>, +

$x^{(3)}$ = <Teh, O, Gai Dai, Peng, Gau, Tabao>, –

$x^{(4)}$ = <Kopi, Si, Ga Dai, Peng, Por, Tabao>, +

h_0 =<∅,∅,∅, ∅,∅,∅>

h_1 = <Kopi, Si, Kosong, Peng, Gau, Lim>

h_2 = <Kopi, Si, ?, Peng, Gau, Lim>

h_3 = <Kopi, Si, ?, Peng, Gau, Lim>

h_4 = <Kopi, Si, ?, Peng, ? ,? >

Limitations of Find-S

- Can't tell whether Find-S has learned target concept
- Can't tell when training examples are inconsistent (i.e., contain errors or noise)
- Picks a maximally specific h (why?)
- Depending on $\mathcal{H}$, there might be several!